

A Unified FIDE, Chess.com, and Lichess Database: Analyzing Elite Chess Players and their Behavioral Patterns

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Abstract

This report introduces a unified database of elite chess players built from three fundamental sources — FIDE, Chess.com, and Lichess — and uses it to study behavioral patterns in online chess. Building this database is the central contribution and the main research challenge of the project: official over-the-board (FIDE) identities and online playing accounts are stored in incompatible formats, exposed through different access channels, and rarely share an explicit key. The motivation for unifying them is to let researchers translate existing “over-the-board” findings to online chess at scale. To demonstrate the value of the database, we pursue two research questions that share the same linked identities but probe different behaviors. The first is *behavioral and social*: do players behave differently (in game length / resignation and opening posture) when facing female opponents, where the opponent’s gender is established through the FIDE-linked identity layer? The second is *strategic*: which factors — player strength, activity, and gender — shape a player’s opening repertoire and its diversity? The two questions are connected by the database itself: the same FIDE-anchored identity map that lets us assign opponent gender also lets us measure each player’s repertoire over their full online game history. We report statistically detectable but small game-length differences by opponent-gender context (and confirm that absolute rating difference, not signed rating difference, is the relevant control), and a strong link between opening diversity and player activity/strength. We present full regression tables, discuss the low explanatory power of the behavioral model, and outline the limitations of perceived-gender inference.

Keywords: data scraping, record linkage, big data, gender effects, opening repertoire, data integration, behavioral modeling, regression analysis.

1 Introduction

There are sports in which players do not start from identical conditions. Chess, by contrast, is usually regarded as exceptionally fair: the only structural asymmetry is that White has a small first-move advantage in the opening phase. Tournament formats try to balance this. In a round-robin event each player receives an (almost) equal number of games with White and Black. In the Swiss system — by far the most common format for open tournaments — colours are balanced as far as possible across a player’s games, but an exactly equal split is not guaranteed. Either way, the rules themselves do not favour any participant.

Equal rules, however, do not imply equal behavioral dynamics. Chess has strong psychological dimensions — expectations about the opponent, opening preparation, and social context —

that influence decision-making during a game and have deeper implications for player behavior. One of the most debated questions in the chess world is whether women-only titles (WGM, WIM, ...) motivate female players or instead constrain their perceived potential [3]. With the rise of online chess and the impact of strong engines on opening theory, the psychology of competitive chess has evolved further over the past decade. Yet academic research has focused almost entirely on “over-the-board” (OTB) chess [1, 2, 4], and online and OTB activity have not been analyzed together in a unified way.

The data-integration research challenge. Closing that gap is not only an analysis task — it is a data-engineering and record-linkage research challenge, and it is the part of this project that required the most effort. FIDE, Chess.com, and Lichess describe the *same* elite players, but they do so through three incompatible representations: FIDE publishes a fixed-width monthly text list keyed by a numeric FIDE ID; Chess.com and Lichess expose JSON APIs keyed by free-form usernames that contain no FIDE ID. There is therefore no shared key, names are formatted differently, and the same person may appear under different spellings, titles, and rating scales on each platform. Constructing a single player entity that is consistent across all three sources — and doing so without falsely merging distinct people — is the core methodological problem we solve in Section 3. We therefore treat the unified database as the project’s primary deliverable and frame the two research questions as demonstrations of what the database makes possible.

Scope. We build a per-player profile that spans the three conditions elite players compete under (FIDE, Chess.com, Lichess) and analyze online games against this profile. The goal is *not* to claim universal causal conclusions about the entire chess population, but to provide transparent, reproducible evidence for the elite online setting and to encourage academic research to incorporate online play.

2 Previous Work and Research Questions

2.1 Previous Work

Several studies document gender differences and stereotype effects in chess. Representative examples include work on participation rates and the over-representation of strong male players [1, 2], the role of gender stereotypes under direct competition [3], and analyses of competition and performance using large rated-game samples [4]. These works are cited here as illustrative examples rather than an exhaustive survey. Crucially, they study OTB play and do not account for the players’ online presence on Chess.com and Lichess, platforms that have only become central in the last decade and remain comparatively under-studied. Our work analyzes online games and links them to official FIDE profiles wherever possible, which makes it feasible to contrast OTB and online behavior. Where an official gender label is unavailable, we use a clearly delimited notion of *perceived gender* (defined in Section 3.5).

2.2 Research Questions

The project targets two research questions that share the unified identity layer.

RQ1 (behavioral / social). *Do players behave differently when facing female opponents compared to male opponents?* We assess this through game-level behavioral signals: game length in plies (a proxy for effort and for how long the losing side continues before resigning) and the aggressiveness of the opening chosen. Move-by-move decision quality (e.g. engine-based ACPL)

is operationally deferred to future work and is discussed as a planned extension; in this version, “decision quality” is captured indirectly through these game-level signals.

RQ2 (strategic). *Which factors drive a player’s opening repertoire?* We quantify two aspects of opening style — aggressiveness and diversity (Shannon entropy of the ECO distribution) — and test whether they are explained by player strength, activity, and gender. The aim is to determine whether opening adaptation is primarily strength/activity driven or gender driven.

3 Data Collection and Unified Database

This section describes the database in detail, because its construction is the project’s main research challenge. We treat FIDE as the source of truth for identity and enrich it with the two largest online platforms.

3.1 Building the FIDE Base Layer

FIDE publishes a complete *Standard Rating List* every month as a downloadable fixed-width text file from its public ratings download area. Each record contains a unique FIDE ID, full name, federation, sex, title(s), and the classical rating, so a single monthly list is effectively an overview of every rated player and is the natural starting point for scraping the titled population — no per-player crawling is required. We parsed the September 2025 standard list, which contains **518,309** rated players, and filtered to titled players to form the base layer. The non-standard fixed-width format (no delimiters; columns defined by character offsets, with optional women’s and other-title fields) was itself a parsing challenge and is the reason FIDE could not simply be queried like an API.

The list contains **23,366** titled players, distributed as shown in Table 1. This number is the denominator for the coverage discussion below.

Table 1: Titled players in the FIDE Standard Rating List (September 2025), by title.

Title	Count	Title	Count
GM (Grandmaster)	1,867	WGM (Woman GM)	335
IM (International Master)	4,224	WIM (Woman IM)	903
FM (FIDE Master)	9,388	WFM (Woman FM)	2,105
CM (Candidate Master)	3,328	WCM (Woman CM)	1,215
Total titled players		23,366	

3.2 Chess.com and Lichess Collection

Both online platforms expose public APIs for titled accounts.

- **Chess.com.** We iterated the official titled-player endpoint (`/pub/titled/{title}`) over all title codes (GM, IM, FM, ...) to obtain the full list of titled usernames ($\approx 15,000$ accounts), then enriched each with the `profile` and `stats` endpoints (threaded) to obtain real name, country, and blitz/rapid/bullet ratings (15,423 enriched profiles).
- **Lichess.** Lichess does not expose a single “all titled players” endpoint, so we combined three sources: members of the `titled-arena` and related teams, the top 200 of every

leaderboard, and — crucially — the participant lists of historic Titled Arena tournaments (2018–2024), which surface *active* titled players. We then bulk-fetched profiles (POST /api/users) for real name, country, and ratings ($\approx 5,400$ accounts; 4,746 enriched profiles).

3.3 Identity Matching and Unification

The platforms share no common key, so linkage is name-based. The unification pipeline (`data/unify_datasets.py`) proceeds as follows:

1. **Name normalization.** FIDE stores names as “Last, First”; platform names are free-form. We normalize both to a canonical lowercase “first last” form, stripping punctuation and diacritics.
2. **Base layer.** FIDE titled players (with their FIDE ID retained as the anchor key) form the primary table.
3. **Matching.** Chess.com and Lichess accounts are matched to FIDE records on the normalized name; title and federation are used to disambiguate near-collisions.
4. **Additions.** Platform-titled players who do not match any FIDE record — for example, holders of national/online titles such as NM (National Master) that do not appear in the FIDE classical list — are added as new rows rather than discarded.

The result is `data/unified_chess_players.csv` with **33,713** unique players. We note that name-based matching is a deliberate, conservative choice: a looser rule would inflate the sample but risk merging distinct people with similar names, so ambiguous matches are kept separate. Replacing name matching with a verified FIDE-ID cross-link (where platforms begin to expose it) is identified as future work.

3.4 Coverage

Because the unified database is anchored on the full monthly FIDE standard list, it covers essentially the *entire* FIDE-titled population of 23,366 players. The unified count (33,713) exceeds this because it additionally includes platform-titled players who are not in the FIDE classical list (step 4 above): online/national titles and accounts not present in the standard rating file. In other words, the database is a superset of the FIDE titled population, extended by the online platforms, rather than a sample of it.

3.5 Gender and Perceived Gender

Gender is assigned by a tiered procedure, defined here (before the variables that use it):

1. **Confirmed.** If a FIDE record exists, we use the FIDE `sex` field (M/F). This is the most reliable layer.
2. **Perceived (inferred).** If no FIDE label is available, we look up the player’s first name in a curated names database; if a gender can be assigned with confidence, the player is marked *perceived* female/male. Ambiguous cases are excluded.
3. **Unknown.** Otherwise the player is marked unknown.

The notion of *perceived gender* reflects how opponent gender is actually available *during* an online game. Although anonymous play is possible, strong players typically reveal a gendered identity through women-only FIDE titles (WGM, WIM) shown next to the account, through real-name usernames, or through profile pictures. Our working assumption is that a single glance at these cues is enough for an opponent to perceive gender before the game begins, which is what RQ1 is designed to probe.

Accordingly, a game is labeled **Male vs Female (Confirmed)** when the opponent’s gender comes from the FIDE-linked mapping, and **Male vs Perceived Female (Inferred)** when it comes from the name-based lookup. A game is treated as inferred if at least one side is not confirmed. Confirmed and inferred cases are kept separate throughout the analysis so that the reader can judge each on its own reliability.

3.6 Dataset Profile

Besides individual player characteristics, the dataset includes more than a million online *games* (not matches: in chess a “match” is a multi-game contest between two players) in which at least one side carries a gender label. Table 2 summarizes the snapshot.

Table 2: Unified dataset snapshot.

Metric	Value
Total unified players	33,713
Male (after inference)	18,716 (55.52%)
Female (after inference)	4,890 (14.50%)
Unknown (after inference)	10,107 (29.98%)
Games with ≥ 1 gender signal	1,042,573
RQ1 analysis subset (games)	25,554

3.7 Cleaning and Missing Values

Cleaning standardized ratings, titles, and federation fields so that a variable has the same meaning across sources, and removed duplicate identity records so that one person is not counted multiple times. Missing values were handled by type. Some are expected and harmless (a platform-specific field that only exists on one source); these are simply ignored where irrelevant. Others are analysis-critical (gender, rating); a record is then used only in the part of the analysis where the required variables are present, so a weak assumption does not propagate into the whole model.

4 Data Analysis

4.1 Variables and Model Inputs

Table 3 lists the variables used repeatedly in the models. The **Level** column indicates whether each variable is defined per *game* (RQ1) or per *player* (RQ2), since the table mixes both.

Table 3: Main variables used in the analysis. “Level” distinguishes game-level (RQ1) from player-level (RQ2) variables.

Variable	Level	Role	Why it matters
<i>PlyCount</i>	Game	DV (RQ1)	Game length; proxy for effort / resignation timing.
<i>IsAggressive</i>	Game	Control / style	Whether the opening is sharp or quiet.
<i>RatingDiff</i>	Game	Control	Signed strength mismatch (White – Black).
$ RatingDiff $	Game	Control (added)	Absolute strength mismatch (see §4.4).
Confirmed-female ind.	Game	Main RQ1 regr.	Tests confirmed female-target games vs baseline.
Perceived-female ind.	Game	Main RQ1 regr.	Keeps inferred group separate from confirmed.
Standardized rating	Player	Predictor (RQ2)	Player strength on a comparable scale.
Gender indicator	Player	Predictor (RQ2)	Whether style relates to player gender.
$\log(\text{games count})$	Player	Predictor (RQ2)	Activity level; logged so extreme counts do not dominate.
Shannon entropy	Player	DV (RQ2)	Breadth of the opening repertoire.

Metric definitions. *Aggressiveness.* For RQ1 (game level) an opening is flagged aggressive using an ECO-based heuristic (codes beginning with B, C, D3, D4, or D5). For RQ2 (player level) aggressiveness is the share of a player’s games whose opening name contains aggressive-keyword markers (Gambit, Attack, Sicilian, King’s Indian). The two definitions differ because the levels of analysis differ; both are stated explicitly so the analyses stay comparable. *Diversity.* Opening diversity is the Shannon entropy over each player’s ECO-code distribution, computed only for active players with at least 10 games as White.

4.2 Model Specifications

Both research questions use ordinary least squares (OLS) regression to isolate a predictor while holding the others constant. **Both models include an intercept** (β_0, α_0); the intercepts are reported in the results tables, so the small R^2 values discussed below reflect genuinely limited explanatory power of the chosen predictors rather than a missing constant term.

RQ1 — opponent gender and game length.

$$\text{PlyCount}_i = \beta_0 + \beta_1 F_i^{\text{conf}} + \beta_2 F_i^{\text{perc}} + \beta_3 (\text{Rating control})_i + \beta_4 \text{IsAggressive}_i + \epsilon_i \quad (1)$$

where F_i^{conf} and F_i^{perc} are the confirmed- and perceived-female indicators (operational dataset categories based on source confidence, not direct observation of intent), and the rating control

is either signed *RatingDiff* or its absolute value $|\text{RatingDiff}|$ (compared in §4.4). A negative coefficient means shorter games.

RQ2 — what drives opening style.

$$\text{Outcome}_i = \alpha_0 + \alpha_1 \text{StdRating}_i + \alpha_2 \text{Gender}_i + \alpha_3 \log(\text{Games}_i) + \eta_i \quad (2)$$

estimated once with aggressiveness and once with diversity as the outcome. Platform was part of the original specification, but in the player-level RQ2 sample every player resolves to a single platform (Lichess; see §4.5), so the platform term is constant and is omitted to avoid collinearity with the intercept.

4.3 Statistical Methods

For RQ1 we use a two-sample *t*-test on game length across opponent-gender contexts, a χ^2 test on the distribution of opening aggressiveness, and the multivariable regression in Eq. (1). For RQ2 we first compute Pearson correlations by gender, then estimate the two regressions in Eq. (2). Results are read on three axes: *direction* (sign), *evidence* (standard error / *p*-value), and *practical size* (coefficient magnitude and R^2). A small *p*-value is not treated as proof of a large effect.

4.4 RQ1 Results: Opponent Gender and Game Length

Descriptive comparison. In the expanded RQ1 sample (25,554 games: 294 Male vs Female confirmed, 8,176 Male vs Perceived Female, 17,084 Male vs Male), female-target contexts have a mean game length of **81.98** plies ($n = 8,470$) versus **85.39** plies ($n = 17,084$) for the male baseline — a difference of -3.41 plies, $t = -7.54$, $p = 4.69 \times 10^{-14}$. Aggressive openings occur in 52.41% of female-target games versus 50.07% of male-baseline games (χ^2 test $p = 7.87 \times 10^{-6}$).

Multivariable regression. Table 4 reports the full regression for Eq. (1), including standard errors, *p*-values, and model diagnostics. Model 1 uses signed *RatingDiff*; Model 2 replaces it with $|\text{RatingDiff}|$, following the reviewer’s observation (Fig. 1(d)) that the rating effect is centered on zero rather than linear.

Table 4: RQ1 regression. Dependent variable: *PlyCount* (game length in plies). Standard errors in parentheses. Model 1 uses signed rating difference; Model 2 uses absolute rating difference. $n = 25,554$.

Predictor	Model 1 (signed)	Model 2 ($ \Delta\text{rating} $)
Intercept (β_0)	86.210*** (0.337)	88.008*** (0.420)
Confirmed-female	-4.378* (2.007)	-3.944* (2.002)
Perceived-female	-3.269*** (0.459)	-2.987*** (0.460)
RatingDiff (signed)	-0.0026 (0.0017)	—
$ \text{RatingDiff} $	—	-0.0193*** (0.0027)
IsAggressive	-1.633*** (0.426)	-1.588*** (0.426)
R^2	0.00289	0.00482
Adjusted R^2	0.00274	0.00466
F statistic	18.52	30.92
$p(F)$	3.3×10^{-15}	1.0×10^{-25}

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Signed *RatingDiff* is not significant ($p = 0.119$); $|\text{RatingDiff}|$ is highly significant ($p = 6.1 \times 10^{-13}$).

Interpretation. Three points follow. (i) After controls, both female-target contexts remain associated with *shorter* games (confirmed ≈ -4 plies, perceived ≈ -3 plies), so the raw difference is not merely an artifact of rating mismatch or opening style. (ii) The reviewer’s point about rating difference is confirmed empirically: the *signed* rating difference has no detectable linear effect ($p = 0.119$), whereas the *absolute* rating difference is highly significant and roughly doubles the explained variance (R^2 : 0.0029 $\rightarrow$ 0.0048). Absolute mismatch shortens games regardless of who is stronger, which matches the zero-centered pattern in Fig. 1(d) and the monotonic decline in Fig. 2(b). (iii) The R^2 remains small. This is expected: game length is driven by many factors absent from the model — time control, clock pressure, specific opening theory depth, individual player tendencies, engine preparation, and event stakes. The intercept (≈ 86 –88 plies) is the baseline game length, so the small R^2 reflects limited predictor coverage, not a missing constant. The effect should therefore be read as a robust *direction* of small practical size, not a complete model of behavior.

RQ1: Opponent gender, opening style, and game length (expanded sample, $n = 25,554$)

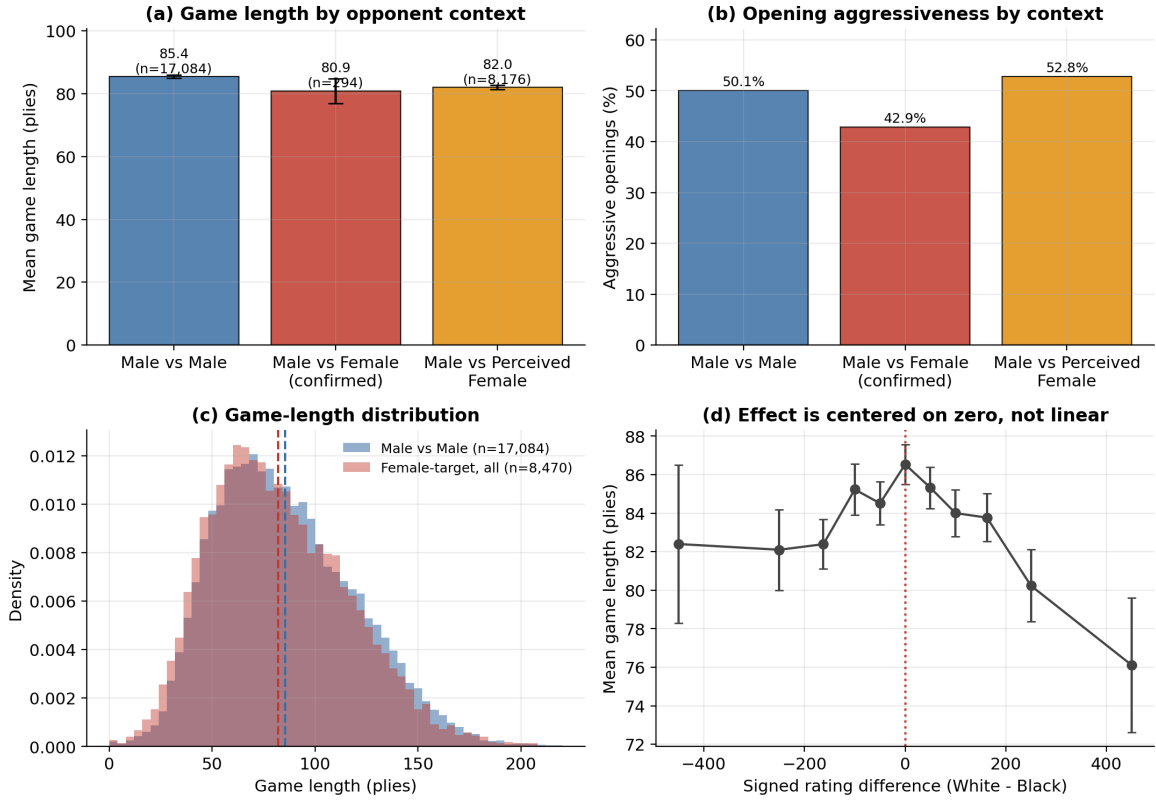


Figure 1: Expanded RQ1 analysis ($n = 25,554$): (a) mean game length by opponent-gender context with 95% confidence intervals; (b) aggressive-opening rate by context; (c) game-length distribution for female-target vs. male-baseline games; (d) mean game length across signed rating-difference bins. Panel (d) shows that the rating effect is centered on zero rather than linear, which motivates using the *absolute* rating difference as a control (Fig. 2).

RQ1: Why absolute rating difference is the right control

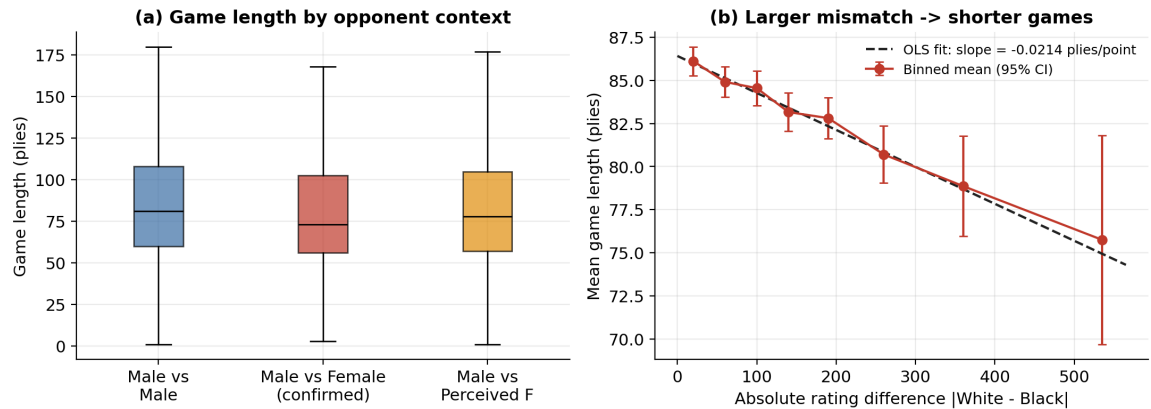


Figure 2: Why the absolute rating difference is the correct control. (a) Game-length distribution (boxplot) by opponent-gender context. (b) Mean game length declines monotonically with the *absolute* rating difference (binned means with 95% CI and an OLS fit), in contrast to the zero-centered pattern against the signed difference in Fig. 1(d).

4.5 RQ2 Results: Rating, Activity, and Opening Style

The player-level sample contains 729 players (700 male, 29 female), all of whom resolve to Lichess in this feature table; platform is therefore constant and omitted from Eq. (2).

Correlations. Diversity correlates positively with rating for both groups (male $r = 0.293$, $p < 0.001$; female $r = 0.620$, $p < 0.001$). Aggressiveness correlates weakly and non-significantly with rating (male $r = -0.072$, $p = 0.057$; female $r = 0.227$, $p = 0.235$).

Multivariable regression. Table 5 reports both models with full diagnostics.

Table 5: RQ2 regressions (player level). Standard errors in parentheses. Predictors: standardized rating, female indicator, and log(games). Platform omitted (constant = Lichess). $n = 729$.

Predictor	Aggressiveness	Diversity (Shannon)
Intercept (α_0)	0.3994*** (0.0185)	1.5488*** (0.0799)
StdRating	-0.0106* (0.0050)	0.1059*** (0.0218)
Female	0.0015 (0.0249)	0.0927 (0.1078)
log(games)	0.0072 (0.0037)	0.3601*** (0.0160)
R^2	0.0089	0.4662
Adjusted R^2	0.0048	0.4640
F statistic	2.16	211.10
$p(F)$	0.092	2.1×10^{-98}

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Interpretation. Diversity is the most conclusive opening-related result: activity (log games) is the dominant driver ($p < 10^{-80}$), with rating a strong secondary driver, and together they explain about 47% of the variance. In both models the *gender* coefficient is small and not statistically significant ($p = 0.95$ for aggressiveness, $p = 0.39$ for diversity), so opening style is shaped by strength and activity rather than by gender. The aggressiveness model explains little variance ($R^2 = 0.009$; overall F not significant), so we treat aggressiveness as a weak, exploratory signal and diversity as the mature RQ2 finding. Two caveats temper the gender comparison: the female subsample is small ($n = 29$), and the higher female diversity–rating correlation should not be over-interpreted given that width.

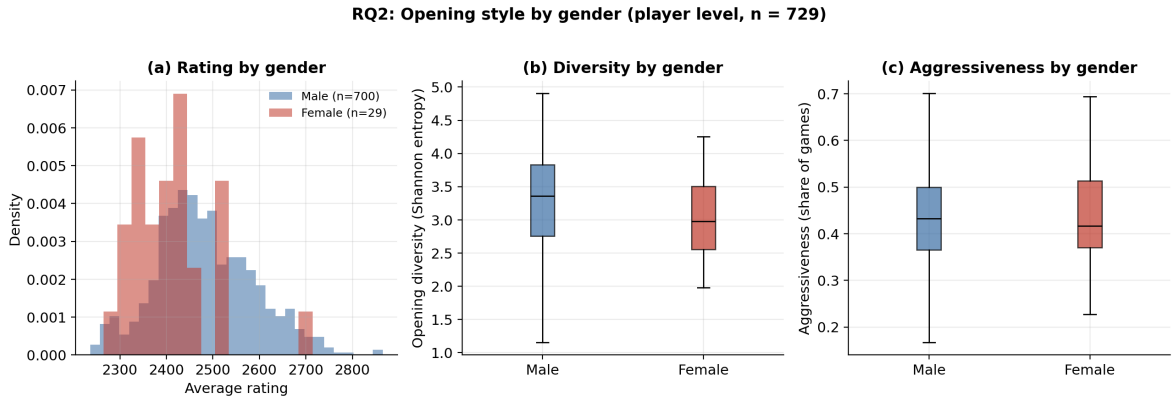


Figure 3: RQ2 opening style by gender (player level, $n = 729$): (a) rating distribution, (b) opening diversity (Shannon entropy), and (c) opening aggressiveness.

RQ2: Opening diversity scales with activity and rating

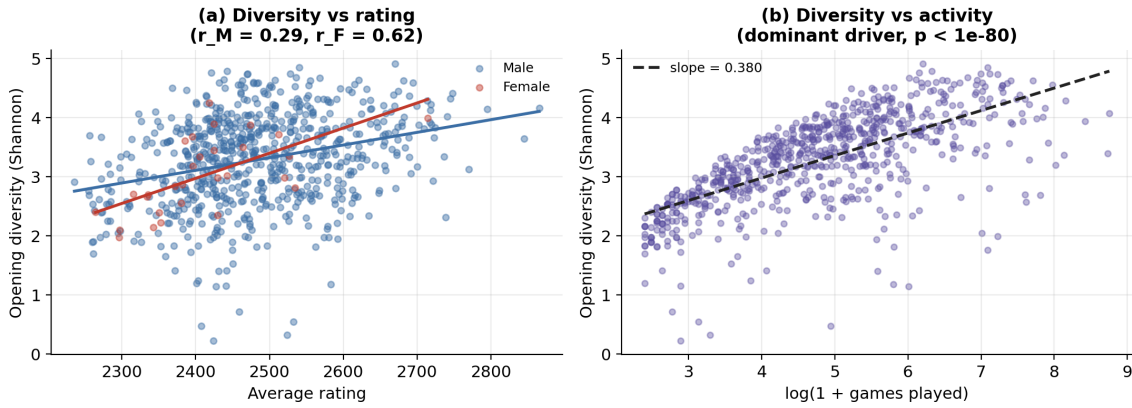


Figure 4: Drivers of opening diversity. (a) Diversity vs. average rating with per-gender linear fits ($r_{\text{male}} = 0.29$, $r_{\text{female}} = 0.62$). (b) Diversity vs. activity (log games), the dominant predictor in the regression ($p < 10^{-80}$).

5 Discussion

What the results say. For RQ1, players do not behave identically against female-target opponents: games are modestly but reliably shorter, with a small shift toward aggressive openings. The effect survives controls for opening style and — importantly — for the *absolute* rating mismatch, which we showed is the correct rating control. For RQ2, opening diversity is strongly tied to a player’s activity and strength, while gender adds no significant explanatory power; opening aggressiveness is only weakly explained by the current feature set.

Why the RQ1 model has low R^2 , and what is missing. A low R^2 with a highly significant coefficient simply means the predictor reliably shifts the mean by a small amount while most of the variance lies elsewhere. Plausible missing predictors include: time control (bullet/blitz/rapid) and per-move clock data; opening-theory depth and whether a line is forcing; player-specific tendencies (fixed effects); engine preparation; rating tier interactions; and game importance. Adding clock and ACPL features is the most promising route to a better-specified behavioral model. We also stress *association, not causation*: these regressions identify conditional correlations, not causal effects.

5.1 Limitations

The database is well suited to elite *online* analysis but is not globally representative of all chess players. Perceived-gender labels are probabilistic and may misclassify; we mitigate this by separating confirmed from perceived groups, but residual error remains. Cross-platform linkage is name-based, which can both miss true matches and, in rare cases, over-merge common names. The RQ2 player sample is Lichess-only and the female subsample is small ($n = 29$). Several models have low explanatory power despite statistical significance, so practical effect sizes must be read alongside p -values.

5.2 Future Work

Planned extensions are: (i) engine-based move-quality (ACPL) and per-phase time-allocation features to give RQ1 a direct decision-quality channel; (ii) FIDE-ID-verified cross-platform linkage to replace name-only matching; (iii) robustness checks across time-control, platform, and rating strata, and across title-based vs. broader-player samples; and (iv) extending the RQ2 feature set so aggressiveness can be modeled as reliably as diversity.

6 Conclusion

The project’s central contribution is a reproducible unified database linking FIDE, Chess.com, and Lichess, which we argue is a research challenge in its own right. Using it, RQ1 shows that female-target contexts correspond to shorter games and a small shift toward aggressive openings, even after controlling for opening style and absolute rating difference; the behavioral direction is robust though small in practical size. RQ2 shows that opening diversity is clearly driven by player activity and strength, with gender playing no significant role, while aggressiveness remains weakly explained. Together these analyses demonstrate that the unified database makes principled, controlled behavioral analysis of elite online chess feasible.

Supplementary Information

Figure Index

All figures in this manuscript are generated from the analysis data by `figures/make_figures.py` and stored in `figures/`:

- `fig1_rq1_overview.png` (Fig. 1)
- `fig2_rq1_rating_control.png` (Fig. 2)
- `fig3_rq2_overview.png` (Fig. 3)
- `fig4_rq2_diversity_drivers.png` (Fig. 4)

Additional exploratory plots from the pipeline are available under `analysis/plots/` and `results/plots/`.

Reproducibility

Core scripts:

- `analysis/05_comprehensive_lichess_extraction.py`
- `analysis/06_rq1_expanded_analysis.py`
- `analysis/02_rq2_opening_repertoire_analysis.py`
- `analysis/07_comprehensive_statistics_report.py`

Key generated outputs:

- `data/rq1_analysis_expanded.csv`
- `analysis/games_male_vs_perceived_female.csv`
- `analysis/rq1_expanded_results.csv`
- `analysis/rq2_regression_results.csv`
- `analysis/comprehensive_statistics_summary.csv`

Repository: <https://github.com/sajayshenoy/Masters-Project>.

References

- [1] M. Bilalić, K. Smallbone, P. McLeod, and F. Gobet, “Why are (the best) women so good at chess? Participation rates and gender differences in intellectual domains,” *Proceedings of the Royal Society B: Biological Sciences*, vol. 276, no. 1659, pp. 1161–1165, 2009.
- [2] A. Blanch, “Sex differences in chess performance: Analyzing participation rates, age, and practice,” *Personality and Individual Differences*, vol. 104, pp. 117–121, 2017.
- [3] A. Maass, C. D’Ettole, and M. Cadinu, “Checkmate? The role of gender stereotypes in the ultimate intellectual sport,” *European Journal of Social Psychology*, vol. 38, no. 2, pp. 231–245, 2008.
- [4] P. Backus, M. Cubel, M. Guid, S. Sánchez-Pagés, and E. López Mañas, “Gender, competition, and performance: Evidence from chess players,” *Quantitative Economics*, vol. 14, no. 1, pp. 1–33, 2023.